

GE261004 DESIGN THINKING

Unit IV – Prototyping and Testing · Presentation 6 of 6

Iterative Design and Improvement

Build, measure, learn · one question per loop · failing fast and cheaply · turning feedback into decisions · when to stop iterating

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In this presentation

- The loop
- One question per loop
- Failing fast and cheaply
- How a loop can end
- From feedback to change
- Deciding what to change
- Four improvement strategies
- When to stop iterating

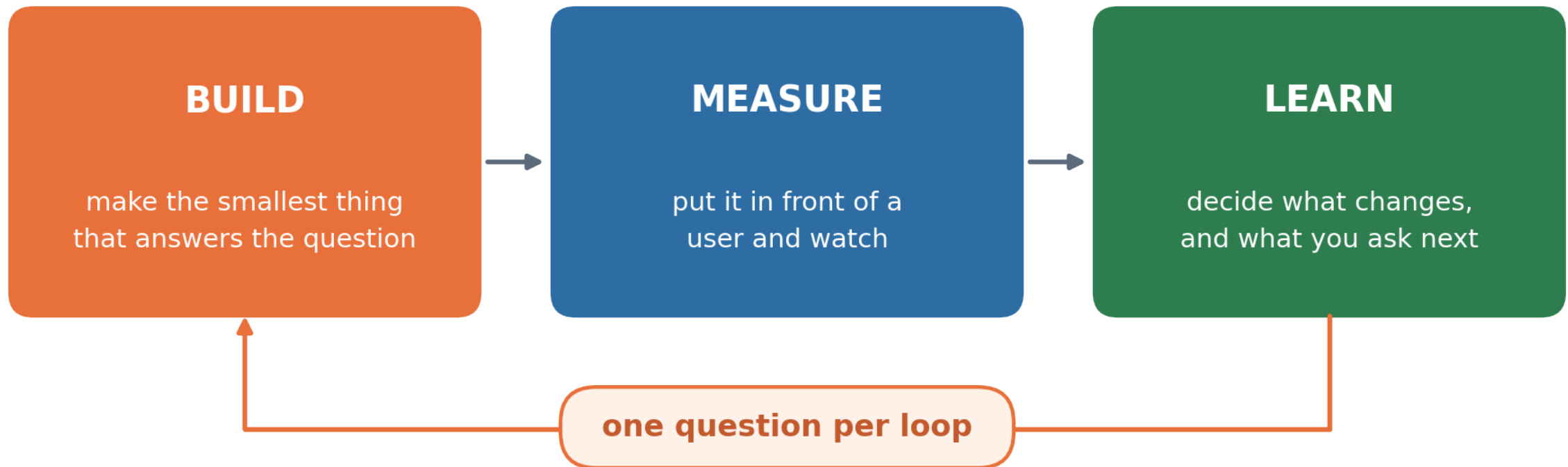


1

The loop

Not a sign that something went wrong – it is the method

Build, measure, learn



the loop is the method — a design that never looped was never tested

Eric Ries named this loop in The Lean Startup, 2011.

One question per loop

- Each loop starts with a question and ends with an answer you can write down
- Build only enough to answer it; everything else is next loop's problem
- Short loops beat big ones – three small loops teach more than one long one
- Record the question, what you built, what happened, and what you decided

KEY POINT A team that cannot say what question this loop answered is not iterating. It is just still building.

Failing fast and cheaply

- “Fail fast” does not mean being careless – it means finding the flaw while it is cheap
- A prototype that fails has done its job; a prototype that cannot fail was not a test
- The expensive failure is the one discovered after tooling, not the one found in cardboard
- Say out loud what would count as failure, before the test, so you cannot reinterpret it after

KEY POINT Cheap failures are how you avoid expensive ones. That is the entire argument for prototyping.

Three ways a loop can end

Persevere

The evidence supports the direction. Refine it and ask a harder question next.

Pivot

The evidence is clear and unfavourable. Keep the insight, change the solution.

Park

The evidence is unclear. Say what would settle it, and go and get that instead.

KEY POINT “We will build it better next time” is none of these three. It is a decision postponed.

2

Improving the design

From a wall of notes to a change you can defend

From feedback to change

- Cluster the grid across users; count how many people hit each problem
- Separate what they said from what you watched them do – and trust the watching
- Rank by how much it hurts, not by how easy it is to fix
- For each change, write the observation that justified it

KEY POINT A change with no observation behind it is a preference. Preferences are fine, but label them.

From finding to decision

What you saw	What it means	What you do
Four of five hesitated	The step is unclear	Redesign that step
One user asked for a feature	One user's wish	Note it; do not build it
Everyone used it wrongly	Your model is wrong	Reframe, do not relabel
It broke in ordinary use	Implementation problem	Fix before testing again

Four ways to improve a design

Remove

Take something out and test again. The cheapest improvement, and the most often skipped.

Reorder

Change the sequence of steps rather than the parts. Often fixes hesitation on its own.

Make it obvious

Do not add a label to explain a confusing shape. Change the shape.

Lower the cost

Less money, less effort, less time, fewer steps – any of the four counts.

When to stop iterating

- When the last two rounds found the same kind of problem, only smaller
- When the remaining problems are cheaper to fix after launch than before it
- When you have run out of questions worth the cost of another loop
- Never because the deadline arrived – that is stopping, not finishing

KEY POINT Stop when new loops stop producing surprises. Until then, every loop is still buying you something.

Iteration traps

What goes wrong

- Changing five things at once
- Iterating without a question
- Polishing instead of testing
- Defending the first design

What to do instead

- One change, then test again
- Write the question first
- Test at the lowest fidelity
- Let the evidence win

KEY POINT Change five things and something improves – but you will never know which of the five did it.

In class: one more loop

- Take the grid from your five tests and cluster it on the table
- Pick the single change that would help the most people
- Make that one change to the prototype, now, in the room
- Test it on two fresh people and write down whether it worked

ACTIVITY Fifty minutes. One change, two users, one sentence: did it work or not?

THINK ABOUT IT

What is the one change your team keeps avoiding because it would mean rebuilding?

- What evidence is pointing at it?
- How long have you known?
- What would it cost to make the change now, against after Unit V?
- What are you actually protecting – the design, or the work you put in?

Summary

- Build, measure, learn – one written question per loop
- Failing fast means finding the flaw while it is still cheap
- Every loop ends in persevere, pivot or park
- Trust what you watched over what you were told
- Rank findings by how much they hurt, not by ease of fixing
- Stop when the loops stop producing surprises

Next: Unit V – Innovation, Entrepreneurship and Sustainability

- Innovation and entrepreneurship
- Value proposition and business model basics
- Sustainable, inclusive and social design
- Pitching, storytelling and design ethics